be called 4K. Just like TV manufacturers went ga-ga over the 3D TVs, expect a barrage of 4K TVs in the next couple of years. Actually there is no fixed resolution as such for 4K. Even panels with a display of 3840x2160 termed as Quad HD are being considered under the 4K segment.

Apart from adding over 5 million more pixels than a regular 1080p panel, the advantage of 4K technology is seen on big theatre screens, and with passive 3D on TV. Current passive 3DTVs from LG and Toshiba use half resolution ie. 1920x540 pixels while viewing in 3D. That resolution will be bumped up to 3840x1080 pixel image per eye. So a higher resolution will certainly help make the 3D effect more immersive.

Opinions on the use of 4K TVs in a home environment are mixed with experts calling it overkill. Consider this, it is quite difficult to differentiate between a 720p and a 1080p panel, so how will it be easy to differentiate between a 1080p and a 4K panel? At a distance of 10 feet, it is very difficult to resolve individual pixels on a 1080p panel. With 1080p panels available to upto 65-inches for home use, for 4K the screen size certainly has to be greater than that to be able to enjoy the content. YouTube has support for 4K videos but according to YouTube company engineer Ramesh Sarukkai, the ideal screen size for 4K videos is 25 feet! Yet another concern is availability of true 4K TV content. But TV manufacturers are going all out to woo consumers.

At CES this year, LG took the curtains off its 84-inch 'ultra-definition' LCD TV. Sony unveiled its 4K projector system earlier this year and is already working on 4K TVs alongwith adding the option to upscale HD content to 4K within their Blu-ray players. Sharp went even a step further, showcasing an 85-inch 8K (7680x4320 reolution) prototype which it plans to out in 2020. Toshiba announced a 4K LED backlit LCD TV with glasses-free 3D (autostereoscopic) panel. Samsung showed a prototype of a 4K TV at CES as well.
-Sharp 8K TV demo http://goo.gl/YvYwz

LG 85-inch 4K TV showcased at CES2012

Better cellphone and tablet Battery technologies

This is one area where the technological innovation hasn't been as great. We are still using the same AA and AAA batteries that we did 20 years ago. Also we are using the same lithium-ion batteries that we did say 10 years ago. Majority of our tablets and cellphones house the Li-ion batteries, so do many cameras, but their usage time will not be beyond a day, in most cases. How can we get batteries to last say a week or months? Let us first get the basic working of a Lithium-ion battery. When in use, the lithium ions travel from the anode via the electrolyte to the cathode. While recharging the battery, the lithium ions travel to the anode from the electrolyte. Current bottleneck involves the amount of charge a battery can carry – higher capacity battery is bigger in size which is not desirable in sleek phones and tablets. Secondly, the recharging time for these Lithium-ion batteries is quite high – in terms of hours. But a group of engineers at Northwestern University, headed by Harold Kung, claim that they have found a way to increase battery capacity by up to 10 times the current capacities (allowing cellphones to run on a single charge for a week!) and reducing the recharging time to just 15 minutes! Traditional batteries use multiple layers of carbon-based graphene sheets as anodes. Six carbon atoms can accommodate one lithium atom, so there is a limit there, as to how much a battery capacity can go upto. Silicon on the other hand can accommodate four lithium atoms for every silicon atom, but silicon has a tendency to expand and contract whilst charging which can get problematic.

The other issue - slower recharging time - can be attributed to the fact that the current anode material involves long and thin (one carbon atom thick) sheets of graphene. In order to completely recharge, the lithium ions must travel from the electrolyte through the graphene sheets and settle down between those sheets. This takes time and there is a sort of accumulation of ions at the sheet edges.

Northwestern University researchers have overcome these two issues. For improving the capacity of the battery, they have used cluster of silicon atoms within the graphene sheets to accommodate more lithium atoms – this also keeps in check the volume changes in silicon as the graphene sheet is flexible. For improving recharging times, the researchers have drilled 10-20 nanometer wide holes in the graphene sheets which will allow the lithium ions to reach the anode faster.

This technology is still three to five years away, but its practical applications are immense. Apart from a future where phones have a week long battery life, these batteries can work wonders when used within an electric car.

Sense My Health

Healthcare is one area where technology is quite pertinent. Specially with an elderly population in a country like ours where it is not always easy for children who may be working or studying, to monitor their ailing parents. Using wireless wearable sensors can come in very handy in such scenarios. The sensors together with processing units, make up what is commonly known as a Wireless Body Area Network (WBAN). These networks employ easy-to-install, low-cost and low-power sensors which should be robust